Chapters 1 & 2 — Short Quiz Review

Skills checklist, note-sheet outline, and a 10-question practice quiz (~30 minutes, no calculator)

Name ______________________ Date ______________________

1. What you must be able to do

check every box before the quiz

- ☐ **§1.1** Clear fractions with an LCD and isolate a variable.
- ☐ **§1.1** Solve a radical equation **and test for extraneous roots**.
- ☐ **§1.1** Solve an equation with the variable in a denominator, stating excluded values.
- ☐ **§1.1** Solve an absolute-value equation by splitting into two cases.
- ☐ **§1.1** Solve a literal equation (a formula) for one of its letters.
- ☐ **§1.2** Solve a system by substitution, including one nonlinear equation.
- ☐ **§1.3** Solve a 2×2 or 3×3 system by elimination.
- ☐ **§1.3** Recognize *no solution* (0 = nonzero) vs. *infinitely many* (0 = 0).
- ☐ **§1.4** Find slope, write point-slope and slope-intercept form, find intercepts.
- ☐ **§1.4** Write parallel and perpendicular lines through a given point.
- ☐ **§1.4** Translate a word problem into a system and solve it.
- ☐ **§2.1** Find a domain: denominators ≠ 0, even radicands ≥ 0.
- ☐ **§2.1** Evaluate a function at a number *and at an expression*.
- ☐ **§2.1** Find a range by completing the square.
- ☐ **§2.2** Apply shifts, stretches, and reflections; track a point through them.
- ☐ **§2.2** Use the vertical line test; classify even / odd / neither.
- ☐ **§2.3** Compute $f(g(x))$ and $g(f(x))$ and know they differ.
- ☐ **§2.3** Find the domain of a composition (both restrictions).
- ☐ **§2.4** Find an inverse by solving $y = f(x)$ for x.
- ☐ **§2.4** Explain one-to-one, the horizontal line test, and reflection across $y = x$.

2. Note-sheet outline

write it yourself — that is the studying

Put these on your note sheet, each with one worked example:

- **Definitions:** function, domain, range, one-to-one, inverse, composition.
- **Slope** $m = \dfrac{y_2 - y_1}{x_2 - x_1}$, **point-slope** $y - y_1 = m(x - x_1)$, parallel: same m; perpendicular: $m_1 m_2 = -1$.
- **Completing the square:** $x^2 + bx = (x + b/2)^2 - (b/2)^2$.
- **Domain rules:** denominator ≠ 0; even root ≥ 0; both at once for a root in a denominator (strict inequality).
- **Inverse recipe:** set $y = f(x)$, solve for x, then rename. Domain of f^{-1} = range of f.
- **Transformations:** inside the function → horizontal and backwards; outside → vertical and as written.

Five mistakes that cost the most points

- Squaring both sides and never checking — **always** test candidates in the *original* equation.
- Writing $f^{-1}(x) = 1/f(x)$. The −1 is notation for "inverse," not an exponent.
- Giving $g(f(x))$ when the problem asked for $f(g(x))$.
- Forgetting the excluded values when you clear a denominator or cross-multiply.
- Shifting the wrong direction: $f(x - 3)$ moves the graph **right**, not left.

3. Practice quiz

10 questions · ~30 minutes · no calculator

Time yourself and work without notes. Then check the key on the last page and re-do anything you missed on fresh paper.

1. Solve for x: $5 - 2(3x - 4) = 4x + 3$

2. Solve $A = P(1 + rt)$ for r.

3. Solve for x: $\sqrt{2x + 7} = x + 2$. *List every candidate and say which survive.*

4. Solve the system: $2x + y = 7$ and $3x - 2y = 7$

5. Find the domain of $f(x) = \dfrac{\sqrt{x + 3}}{x - 1}$

6. Let $f(x) = x^2 - 3x$. Find $f(-2)$ and simplify $f(t + 1)$.

7. Let $f(x) = 3x + 2$ and $g(x) = x^2 - 1$. Find $f(g(x))$, $g(f(x))$, and $f(g(-1))$.

8. Find $f^{-1}(x)$ for $f(x) = 5 - 2x^3$.

9. Write the equation of the line through $(1, -4)$ that is parallel to $2x + y = 9$.

10. The graph of $y = f(x)$ contains the point $(-3, 7)$. Find a point that must lie on $y = 2f(x + 1) - 5$.

4. Answer key

check only after you finish

#	Answer	If you missed it, review…
1	$x = 1$	§1.1 — distribute the −2 across *both* terms.
2	$r = \dfrac{A - P}{Pt}$	§1.1 — literal equations; divide by P first, then subtract 1.
3	$x = 1$ only ($x = -3$ is extraneous)	§1.1 — squaring gives $x^2 + 2x - 3 = 0$; both roots must be checked.
4	$(x, y) = (3, 1)$	§1.2–1.3 — substitute $y = 7 - 2x$, or double the first and add.
5	$x \geq -3$ and $x \neq 1$: $[-3, 1) \cup (1, \infty)$	§2.1 — two restrictions apply, not one.
6	$f(-2) = 10$; $f(t + 1) = t^2 - t - 2$	§2.1 — substitute the whole expression, then expand carefully.
7	$3x^2 - 1$; $9x^2 + 12x + 3$; $f(g(-1)) = 2$	§2.3 — inner function first; the two compositions are different.
8	$f^{-1}(x) = \sqrt[3]{\dfrac{5 - x}{2}}$	§2.4 — solve $y = 5 - 2x^3$ for x, then rename.
9	$y = -2x - 2$	§1.4 — the given line has slope −2; parallel lines share it.
10	$(-4, 9)$	§2.2 — solve $x + 1 = -3$ for the input; apply $2(7) - 5$ to the output.